COURSE INFORMATION

Faculty:	Faculty of Computing	**Page:**	1 of 8
Program name:	Bachelor in Software Engineering with Honours		
Course code:	SCSE1224	**Academic Session/Semester:**	2024-2025/2
Course name:	Advanced Programming	**Pre/co requisite (course name and code, if applicable):**	SCSE1013 Fundamental Programming Concept
Credit hours:	4		

Course synopsis	This course presents the advanced programming techniques and features. The course will cover concepts in Object-Oriented Programming and introduce functional programming paradigm. Basic understanding on control structures, objects and classes are required to enrol in this course. The course will also cover some advanced programming techniques including asynchronous programming. The course will equip the students with the theory and practice on problem solving using such techniques. The course will also provide the students with written and oral communication skills. At the end of this course, students should be able to use appropriate programming techniques and tools to develop programs to solve problems.			
Course coordinator	Dr. Mohd Razak bin Samingan			
Course lecturer(s)	**Name**	**Section**	**Contact no.**	**E-mail (@utm.my)**
	Dr. Mohd Razak bin Samingan	01		mrazak
	Mr. Norizam bin Katmon	02		rizam
	Dr. Mohd Razak bin Samingan	03		mrazak
	Dr. Ruhaidah binti Samsudin	04		ruhaidah
	Dr. Liyana 'Adilla Binti Burhanuddin	MJIT		liyanaadilla

Mapping of the Course Learning Outcomes (CLO) to the Programme Learning Outcomes (PLO), Teaching & Learning (T&L) methods and Assessment methods:

No.	CLO	PLO (Code)	*Taxonomies and **generic skills	T&L methods	***Assessment methods
CLO1	Construct programming solution using advanced programming techniques in given time frame.	PLO3 (PS)	C5, P4	Pair Programming, Jigsaw Puzzle	L, SBT, F
CLO2	Communicate solutions to the identified problems in writing and orally.	PLO5 (CS)	CS1, CS2	Project-based	PR, PD

Prepared by:

Name: Dr. Ruhaidah Samsudin

Signature:

Date:

Certified by:

Name: PM Dr. Radziah Mohamad

Signature:

Date:

CLO3	Evaluate advanced programming features and techniques as well as suitable tools to develop solutions to the identified problems.	PLO6 (DS)	C6, P5	Project-based	PR, PD

*** L-Lab Exercise; SBT –Skill-based Test; F – Final Exam; PR-Project Report, PD-Project Demo

Details on Innovative T&L practices:

No.	Type	Implementation
1.	Pair Programming	Conducted through in-class activities. Students will do programming in pairs. Each student will play the role of the driver and navigator interchangeably. Driver will write the code and the Navigator acts like a co-pilot which assists the driver.
2.	Jigsaw Puzzle	Conducted as part of the in-class activities. Students are divided into groups (4 to 6 members per group) called home groups. Each member from each group will be given a topic (topics) to be explored. Students will then be put into their expert groups based on their topics. They will do active discussion in exploring the topics and later will share the finding with their members of the home groups.
3.	Project-based learning	Conducted through a group project (4 students per group). The project is divided into several deliverables, including proposal, interim progress and project finale.

Weekly Schedule:

Week 1 17 - 23 Mar	**Chapter 1 Introduction to Object-oriented concepts and a general overview of Java API** ● Object oriented definition ● Classes in Java ● Java Basics ● Creating Java Programs ● Java Variables and data types
Week 2 24 – 30 Mar	**Chapter 2: Introduction to Classes and Objects** ● Instance Fields, Accessor and Mutator Methods ● Constructors ● UML class diagram ● Static Class Members ● Passing and returning Objects to/from Methods ● Packages and import Statements

Week 3 31 Mar – 6 Apr Hari Raya / Eid-Mubarak (Mon/Tue)	**Chapter 3: Arrays** Defining One-Dimensional Array ● Passing Arrays as Arguments to Methods ● Returning Arrays from Methods ● String Arrays ● Arrays of Objects ● Defining Two-Dimensional Array **Lab Exercise 1**
Week 4 7 – 13 Apr	**Chapter 4: Vectors & Collections** ● Array Lists ● Vector ● File Operations **Project Proposal (Synopsis, System Requirements, Class Descriptions)**
Week 5 14 – 20 Apr	**Chapter 5: Class Relationships** ● Association ● Aggregation ● Composition **Project proposal evaluation**
Week 6 21 – 27 Apr	**Chapter 6: Inheritance** ● Introduction to Inheritance ● Protected Members ● The Object Class ● Calling the Superclass Constructor ● Chains of Inheritance ● Overriding Superclass Methods **Lab Exercise 2**
Week 7 28 Apr – 4 May Labour Day	**Chapter 7: Polymorphism** ● Introduction to Polymorphism ● Abstract Classes and Abstract Methods ● Interface and implements

(1 May – Thu)	
Week 8 5 – 11 May	**MID-SEMESTER BREAK**
Week 9 12 – 18 May Wesak Day (12 May – Mon)	**Chapter 8: Error Handling** Bugs and errors, debugging, error propagation, exceptions handling: try-catch, **Project Interim Progress (Class Design/Relationships & Working Functions)**
Week 10 19 – 25 May	**Chapter 9: Functional Programming** ● Introduction: what and why? ● Concepts: Functions, Immutability, Side Effect Free, Declarative vs Imperative Programming, Function Composition: High-order functions, Currying ● Closure, Recursion, Arrow Functions, Function Binding, Functions as parameters, Generator Functions, Arrays Map and High-order functions **Lab Exercise 3**
Week 11 26 May – 1 Jun	**Chapter 10: Java Stream** Introduction, Java Stream Benefits, Stream Operations (create, intermediate, terminal), Conversion to Array/List **Skill-Based Test (28 May 2025, Wednesday, 8pm-10pm)**
Week 12 2 – 8 Jun YDPA Birthday (2 Jun – Mon) Hari Raya Haji (8 Jun – Sun)	**Chapter 11: Asynchronous Programming** Introduction: what and why, Event-driven architecture, event loop, Promise, Async Functions, Async-Await syntax, Async loops
Week 13 9 – 15 Jun	**Chapter 12: Regular Expressions** Introduction to Regex: what and why, regex syntax, sets of characters, repeating pattern, grouping sub-expression, pattern boundaries, choice patterns, backtracking, replacing pattern

	Lab Exercise 4
Week 14 16 – 22 Jun	**Course Revision & Test Preparation**
Week 15 23 – 29 Jun	**Demo Day:** Project Finale, Project Demo

Transferable skills (generic skills learned in course of study which can be useful and utilised in other settings):

Communication skills

Student learning time (SLT) details:

Distribution of student Learning Time (SLT) Course content outline					Teaching and Learning Activities		TOTAL SLT
	Guided Learning (Face to Face)				Guided Learning Non-Face to Face	Independent Learning Non-Face to face	
CLO	**L**	**T**	**P**	**O**			
CLO1	12h		7h		12h	32h	**63h**
CLO2				10h		20h	**30h**
CLO3	12h	7h	8h		8h	12h	**47h**
Total SLT	**24h**	**7h**	**15h**	**10h**	**20h**	**60h**	**140h**

L: Lecture, T: Tutorial, P: Practical, O: Others

	Continuous Assessment	PLO	Percentage	Total SLT
1	Lab Exercise (4)	PLO3 (PS)	20	**5h**
2	Project Proposal	PLO5 (CS)	5	**As in CLO2 (2h)**

3	Project Interim Progress	PLO5 (CS) PLO6 (DS)	2 8	**As in CLO2 (1h)** **3h**
4	Project Finale	PLO5 (DS) PLO6 (DS)	3 17	**As in CLO2 (1h)** **6h**
5	Skill-Based Test	PLO3 (PS)	15	**2h**
Final Assessment			Percentage	Total SLT
6	Final Exam	PLO3 (PS)	30	**4h**
Grand Total SLT				**160h**

*To pass this course, students must pass both Continuous and Final assessment.

Special requirement to deliver the course (e.g: software, nursery, computer lab, simulation room):

Computer Lab, Integrated Development Environment (IDE)

Download Links:

1. Software, JAVA SDK (latest version) - https://www.oracle.com/my/java/technologies/downloads/
2. Editor - https://code.visualstudio.com/download
3. UML Diagram Tool - https://plantuml.com/
4. Draw IO - https://app.diagrams.net/

References:

1. Norazah Yusof, Radziah Mohamad, dan Nor Bahiah Hj. Ahmad. Object Oriented Programming Using Java.7th Edition. 2014. Penerbit UTM.
2. Y. Daniel Liang, Introduction to Java programming. 8th edition. 2011. Pearson - Prentice Hall.
3. Deitel, Deitel, Java How to Program, 7th edition. 2007. Pearson – Prentice Hall.
4. Tony Gaddis, Starting out with Java – From Control Structures through Objects, 3rd edition. 2008. Addison Wesley.
5. Walter Savitch, Absolute Java, 2nd edition. 2006. Pearson Addison-Wesley.
6. Joyce Farrel, Java Programming. 4th edition. 2008. Thomson Course Technology.

Online

http://elearning.utm.my

Academic honesty and plagiarism: *(Below is just a sample)*

Assignments are individual tasks and NOT group activities (UNLESS EXPLICITLY INDICATED AS GROUP ACTIVITIES) Copying of work (texts, simulation results etc.) from other students/groups or from other sources is not allowed. Brief quotations are allowed and then only if indicated as such. Existing texts should be reformulated with your own words used to explain what you have read. It is not acceptable to retype existing texts and just acknowledge the source as a reference. Be warned: students who submit copied work will obtain a mark of **zero** for the assignment and disciplinary steps may be taken by the faculty. It is also unacceptable to do somebody else's work, to lend your work to them or to make your work available to them to copy.

Other additional information (Course policy, any specific instruction etc.):

To pass this course, students must fulfil all the following conditions:

1. Get >= 30% of the overall continuous assessment marks (21 / 70)
2. Get >= 20% of the final assessment marks (6 / 30)
3. Get >= 40% for the total of the two above (continuous assessment + final assessment)

** Total mark will be reduced to below UTM's passing mark if not fulfill the above requirements.*

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